

# HART Communication Protocol in Industrial Process Automation: Architecture, Evolution, and Applications

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**Abstract**—The Highway Addressable Remote Transducer (HART) protocol remains the dominant field communication standard in process automation, with an estimated 80-90% incorporating HART capability. [5] Its dual-channel architecture carries the 4-20 mA analog primary variable alongside a 1200 baud Bell 202 FSK digital signal on existing wiring, enabling diagnostics, configuration, and multi-variable access without rewiring. [2] This paper analyses HART within a closed-loop chemical reactor scenario comprising a temperature transmitter (TT-201), a Coriolis flow transmitter (FT-202), and a smart valve positioner (FCV-203) integrated into a DCS PID loop. Three operational scenarios are examined: normal PID operation, remote reconfiguration via HART Command 35, and predictive fault detection through valve positioner diagnostics. A structured comparison against FOUNDATION Fieldbus H1 and PROFIBUS PA is presented, confirming HART unique fitness for brownfield process automation modernisation.

## I. INTRODUCTION

Industrial process plants rely on thousands of field instruments communicating process data to centralised control systems. Historically, this depended exclusively on the 4-20 mA current loop: robust and noise-immune, but limited to one variable per device with no diagnostics or remote configuration. [2] HART, developed by Rosemount in the mid-1980s and later governed by FieldComm Group, resolves this by superimposing a Bell 202 FSK digital signal onto the existing loop. [7] The FSK signal uses 1200 Hz for logic "1" and 2200 Hz for logic "0" at 1200 baud, maintaining a zero mean so the analog DC current remains undisturbed. [7] The result is two simultaneous channels: the analog path for real-time PID control input, and the digital path for configuration, diagnostics, and secondary variable access. [1] Over 60% via HART, [5] representing a massive installed base that operators

prefer to leverage rather than replace with fully digital fieldbus, which requires new wiring and carries higher migration costs. [8] This paper analyses HART in a representative chemical reactor loop, evaluates three practical scenarios, and positions HART relative to competing protocols.

## II. BACKGROUND AND LITERATURE REVIEW

### A. Protocol Evolution

HART 5 established the FSK physical layer and three-tier command set (universal, common practice, device-specific). HART 6 added digital control valves and PID controller device families. HART 7, ratified as IEC 62591, introduced WirelessHART as an exclusively HART 7 feature, providing IEEE 802.15.4 mesh networking at 2.4 GHz with TDMA scheduling, channel hopping, and spread-spectrum security. [3][12] Each version extended the certified device library maintained by FieldComm Group through its device integration centres. [17]

### B. Multi-Variable Capability and Related Work

HART structures device data as Primary (PV), Secondary (SV), Tertiary (TV), and Quaternary (QV) variables. A Coriolis meter, for instance, can deliver mass flow, volumetric flow, temperature, and density over a single cable pair. [1][4] Research has compared HART, PROFIBUS PA, and FOUNDATION Fieldbus on throughput, diagnostics, and installation economics, consistently identifying HART backward compatibility as the key advantage for retrofit applications while noting the 1200 baud digital bandwidth constraint. [18] Studies on WirelessHART have addressed mesh reliability and security. [9] Work on HART-IP (standardised 2012, HART 7.9) demonstrates encapsulation of HART data in IP packets for Ethernet and cloud delivery. [10][11] Digital twin frameworks consuming HART device data for predictive maintenance have

been proposed as a path from device diagnostics to plant-wide asset health management. [16]

### III. SYSTEM ARCHITECTURE AND HART ROLE

The scenario is a temperature-controlled chemical reactor with three HART field devices wired to a DCS (e.g., ABB 800xA or Emerson DeltaV) executing a cascade PID strategy: [17] TT-201: HART temperature transmitter on the reactor vessel. PV (4-20 mA) = reactor temperature. SV (digital) = sensor diagnostic status. FT-202: HART Coriolis flow transmitter on the feed line. PV = mass flow rate. SV, TV, QV = volumetric flow, fluid temperature, and density on a single cable pair. [1] FCV-203: Pneumatic valve with HART smart positioner. DCS set-point via 4-20 mA; positioner reports actual position, supply pressure, and actuator pressure as SV/TV/QV digitally. Diagnostic data include total travel, drive signal deviation, and partial stroke test results. [13][15] Each device connects via dedicated two-wire twisted pair in point-to-point mode. The DCS interfaces through remote I/O modules or HART multiplexers in marshalling cabinets, polling the digital channel sequentially while the analog channel runs uninterrupted. [19] A minimum 230 Ohm loop resistance is required for FSK voltage detection; 250 Ohm is standard practice. [4] The PID function blocks in the DCS sample the 4-20 mA analog inputs at 100-500 ms scan rates. The digital channel operates asynchronously, delivering secondary variables and diagnostics at multiplexer polling rates typically in the range of one to several seconds per device. [18] This preserves control-loop integrity regardless of digital channel activity.

### IV. EVALUATION

#### A. Scenario 1 – Normal PID Operation

During steady operation, TT-201 and FT-202 deliver PV values on the analog channel to the temperature and flow PID controllers respectively. FCV-203 receives its set-point via analog output. All control-path signals are continuous and deterministic. [2][7] Simultaneously, the DCS multiplexer retrieves FT-202 volumetric flow, fluid temperature, and density as SV/TV/QV without additional cabling, [1] and reads FCV-203 actual valve position and actuator pressure, enabling the DCS historian to build a valve travel accumulator for maintenance scheduling. [13]

#### B. Scenario 2 – Remote Reconfiguration

A process change requires TT-201 span update from 0-200 C to 0-300 C. The control room engineer issues HART Command 35 (Write PV Range) through the DCS asset management interface without dispatching a field technician. [4][1] TT201 updates its span and recomputes 4-20 mA scaling. The engineer verifies the update via HART Command 15 (Read PV Range). FieldComm Group operational case studies from chemical and refining plants document significant reductions in maintenance labour hours attributable to remote HART configurability of this type. [5]

TABLE I. Comparison of Key Protocol Attributes

| Attribute                | HART 7                      | FOUNDATION FF H1            | PROFIBUS PA                 |
|--------------------------|-----------------------------|-----------------------------|-----------------------------|
| Signal type              | Hybrid 4-20 mA + FSK        | All-digital IEC 61158-2     | All-digital IEC 61158-2     |
| Digital data rate        | 1200 baud                   | 31.25 kbit/s                | 31.25 kbit/s                |
| Backward compatible      | Yes (reuses 4-20 mA wiring) | No -- new bus wiring needed | No -- new bus wiring needed |
| Control in field device  | No                          | Yes (function blocks)       | No                          |
| Multi-variable per cable | Yes (PV-QV)                 | Yes                         | Yes                         |
| Wireless extension       | WirelessHART IEC 62591      | None native                 | None native                 |
| Best fit                 | Brownfield retrofit, SIS    | Greenfield process control  | Greenfield, Siemens DCS     |

Sources: [18][19]

Fig. 1. Sources: [18][19]

#### C. Scenario 3 – Fault Detection via Valve Diagnostics

During a routine polling cycle, FCV-203 returns a HART status byte indicating elevated drive signal deviation – a condition in which control effort to reach the commanded position has increased beyond threshold, consistent with increasing packing friction. [13][14] A partial stroke test (PST) is initiated remotely via the HART digital channel. The positioner drives the valve to a defined intermediate position and back, and the DCS records position versus actuator pressure. The resulting signature confirms elevated friction in the lower 20% preventing an unplanned process shutdown. HART-enabled PST and continuous positioner diagnostics have been documented as significant contributors to SIL compliance in safety instrumented systems. [14][15]c

#### D. Protocol Comparison

Table I compares HART 7, FOUNDATION Fieldbus H1, and PROFIBUS PA across key process automation attributes. HART key strengths are zero rewiring cost, open multi-vendor interoperability (enforced through FieldComm Group certification), NAMUR NE107-aligned device diagnostics, and operability in SIL-rated safety loops where the analog control path is unaffected by digital diagnostic activity. [14][15] Limitations include the 1200 baud constraint, which precludes hard realtime digital control, and the half-duplex polling model, which serialises data retrieval across multiplexed devices. Multidrop mode, which supports up to 15 devices per loop, fixes the analog current at 4 mA and therefore cannot serve standard PID applications. [1][4]

$$a + b = \gamma \quad (1)$$

### CONCLUSIONS

This paper has demonstrated HART role in a closed-loop chemical reactor scenario covering real-time PID control, remote device reconfiguration, and predictive fault detection. The analysis confirms that the dual-channel architecture provides a practical bridge between established 4-20 mA infrastructure and the diagnostic and configuration capabilities of digital instrumentation, at far lower migration cost than greenfield fieldbus deployment. [5][8] Correct HART configuration within DCS platforms unlocks multi-variable data access and continuous diagnostic visibility, enabling advanced analytics and smarter maintenance strategies. [16][17] The comparison against FOUNDATION Fieldbus H1 and PROFIBUS PA positions HART as the preferred protocol for brownfield retrofit

and safety system applications, where backward compatibility outweighs the higher throughput of fully digital alternatives. [18] The HART ecosystem trajectory supports continued relevance. WirelessHART extends HART to battery-powered mesh sensors without new cabling. [9][12] HART-IP encapsulates HART commands in IP packets, enabling HART device data to reach cloud analytics and IIoT platforms directly from multiplexers and WirelessHART gateways. [10][20] As process industries adopt Industry 4.0 architectures incorporating edge computing, digital twins, and machine-learning-based predictive maintenance, [16] the HART ecosystem offers an evolutionary path that builds on – rather than replaces – the vast installed base of existing field devices.

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